AMATH 515: IMAGE RECOVERY

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1. Introduction

In this paper, I am comparing methods for recovering a corrupted image. The sparsity of an image in the pixel space is the key to recovering the image from corrupt information. To achieve this, I am using the discrete cosine transform (DCT), which introduces sparsity in the images. The image is reconstructed by minimizing a L-1 regression function. I use the CVXPY package to minimize the function and also with a custom implementation of proximal gradient descent.

2. Theory

2.1. **Mathematical Derivations.** In this section, I solve the theoretical equations required to implement the proximal gradient descent algorithm.
1. The proximal operator for L-1 norm can be obtained by solving the minimization problem,

$$prox_{\lambda\|.\|_1}(x) = \arg\min_z(\frac{1}{2}\|z-x\|^2 + \lambda\|z\|_1)$$

which is equivalent to minimizing

$$f(z) = \frac{1}{2}\Sigma_i(z_i - x_i)^2 + \lambda\Sigma_i|z_i|$$

The L1 norm introduces a non differentiable point at $z_i = 0$.
So we consider the following cases, case i: $z_i > 0$, implies $|z_i| = z_i$

$$\begin{aligned}\frac{d}{dz_i}(\frac{1}{2}(z_i - x_i)^2 + \lambda z_i) &= 0\\ z_i - x_i + \lambda &= 0\\ z_i &= x_i - \lambda\end{aligned}$$

For the above equation to be valid we need,

$$\begin{aligned}x_i - \lambda &> 0\\ x_i &> \lambda\end{aligned}$$

case ii: $z_i < 0$, implies $|z_i| = -z_i$

$$\begin{aligned}\frac{d}{dz_i}(\frac{1}{2}(z_i - x_i)^2 + \lambda z_i) &= 0\\ z_i - x_i - \lambda &= 0\\ z_i &= x_i + \lambda\end{aligned}$$

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For the above equation to be valid we need,

$$\begin{aligned} x_i + \lambda &< 0 \\ x_i &< -\lambda \end{aligned}$$

case iii. $z_i = 0$ Here we rely on the sub-gradient, the sub-differential of $|z_i|$ is given by

$$\frac{\partial |z_i|}{\partial z_i} = \begin{cases} 1, & z_i > 0, \\ [-1, 1], & z_i = 0, \\ -1, & z_i < 0. \end{cases}$$

For z_i to be a minimizer,

$$\begin{aligned} 0 \in \partial f(z_i) &= (z_i - x_i) + \lambda |z_i| \\ 0 &\in -x_i + \lambda \cdot [-1, 1] \\ x_i &\in [-\lambda, \lambda] \end{aligned}$$

Summarizing the three case,

$$\begin{cases} x_i - \lambda, & x_i > \lambda, \\ 0, & |x_i| \leq \lambda, \\ x_i + \lambda, & x_i < -\lambda, \end{cases}$$

This is the soft thresholding function given by

$$prox_{\lambda \|.\|_1}(x) = sign(x) \max(|x| - \lambda, 0)$$

2.

$$f(X) = \frac{1}{2} \|M \circ (iDCT(X) - Y)\|_F^2$$

Writing the squared Frobenius norm as summation,

$$f(X) = \frac{1}{2} \sum_{i,j} M_{i,j}^2 (iDCT(X)_{i,j} - Y_{i,j})^2$$

Since the Mask matrix has only 0's and 1's, we can write above summation as,

$$f(X) = \frac{1}{2} \sum_{i,j} M_{i,j} (iDCT(X)_{i,j} - Y_{i,j})^2$$

To take the gradient, we apply chain rule to the expression in the summation.
Let,

$$\begin{aligned} E = M \circ (iDCT(X) - Y) \\ f(X) = \frac{1}{2} \|E\|_F^2 \\ \nabla f(X) = \nabla_X \frac{1}{2} \|E\|_F^2 \\ \nabla f(X) = E \cdot \nabla_X E \end{aligned}$$

To calculate $\nabla_X E$

$$\begin{aligned} \nabla_X E = \nabla_X (M \circ (iDCT(X) - Y)) \\ \nabla_X E = M \circ \nabla_X (iDCT(X) - Y) \\ \nabla_X (iDCT(X) - Y) = A^{-1} \\ \nabla_X E = M \circ A^{-1} \end{aligned}$$

Substituting the result back into the calculation of gradient,

$$\nabla f(X) = (M \circ (iDCT(X) - Y)) \cdot M \circ A^{-1}$$

As M is a diagonal matrix of 0's and 1's, the 2 M's in above equation would combine to give 1 M.

$$\nabla f(X) = (M \circ (iDCT(X) - Y)) \cdot A^{-1}$$

By adjoint property of DCT,

$$< A^{-1}Y, X >_F = < Y, AX >_F$$

applying above property to our expression of gradient,

$$\nabla f(X) = A(M \circ (iDCT(X) - Y))$$

which is the required expression for the gradient.

3. To find Lipschitz constant, L for ∇f,

$$\|\nabla f(X) - \nabla f(X')\|_F \leq L\|X - X'\|_F, \forall X, X'$$

$$f(X) = \frac{1}{2}\|M \circ (iDCT(X) - Y)\|_F$$
$$\nabla f(X) = A(M \circ (iDCT(X) - Y))$$
$$\nabla^2 f(X) = \nabla_X(\nabla f(X))$$
$$\nabla^2 f(X) = A \circ M \circ \nabla_X(iDCT(X))$$
$$\nabla^2 f(X) = A(M \circ A^{-1})$$

The Lipschtiz constant, L is given by the spectral norm of the Hessian.

$$L = \|\nabla^2 f(X)\|_2 = \|A(M \circ A^{-1})\|_2$$

Since A is unitary, its spectral norm is 1. Since M is diagonal matrix with 0 or 1, its spectral norm is at most 1.
Therefore,

$$\|\nabla^2 f(X)\|_2 = \|A(M \circ A^{-1})\|_2 \leq 1$$

This means the $\nabla f(X)$ is 1-Lipschtiz.

2.2. **Computational Methods.**

2.2.1. *Optimization using Library.* In this method, I use the CVXPY library to minimize the objective function. I pre-compute the coefficients of the DCT and iDCT and perform matrix multiplication with the optimization variable. The built-in *minimize* and *solve* methods, return the minimizer.

2.2.2. *Objective components.* This function factory builds and returns the functions required to perform proximal gradient descent. The returned values include the objective function, its gradient, the L-1 regularization function, and the proximal operator.

2.2.3. *PGD solver.* This function performs the minimization. The function takes the DCT of the masked image as the input and performs descent by using the objective components factory. The reduced value is then passed to the proximal operator and next iteration of optimization variable is obtained. The solver performs this until the norm between two consecutive steps is less than tolerance.

2.2.4. *Evaluation criteria.* We use two evaluation criterias to measure the performance of the algorithms. The Mean Squared Errors (MSE) calculates the difference in the mean of masked out pixels between original and the reconstructed image and divides by the variance of same pixels. The Peak Signal to Noise Ratio (PSNR) computes the MSE and normalizes using the peak intensity. It is then converted to logarithmic scale to make it a ratio based interpretation.

3. Results

3.1. **Experiment parameters.** The original, down sampled and masked imaged are displayed in 1

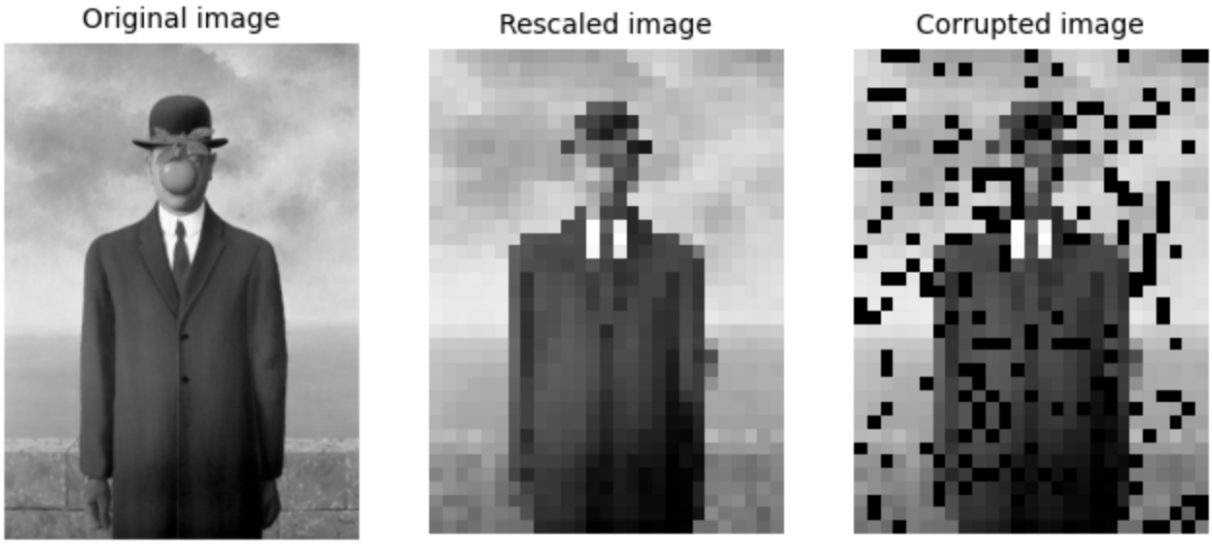


Figure 1. Input images

3.2. **Optimization using library.** The CVXPY library works with a masked version of the down sampled image. The algorithm is relatively slower as it performs the matrix-vector multiplication of DCT coefficients with the optimization variable at every step. The reconstruction image is visibly convincing.

3.3. **PGD Optimizer.** The optimizer was tested on the masked version of down sampled and the original image. The algorithm converged quickly on both versions and the reconstructed images were close to the original. The results of both CVX and PGD from down sampled image is displayed in 2.

3.4. **Parameter tuning.**

3.4.1. *Regularization strength.* This is denoted by λ in the derived equations. This acts as a measure of the details retained by the masked image. The lower regularization value will try to match the masked image exactly. This was verified by lowering the value of λ. The generated images are captured in 3. Note with $\lambda = 1e^{-4}$, the mask appears to be distinctly visible over the corrupted image.

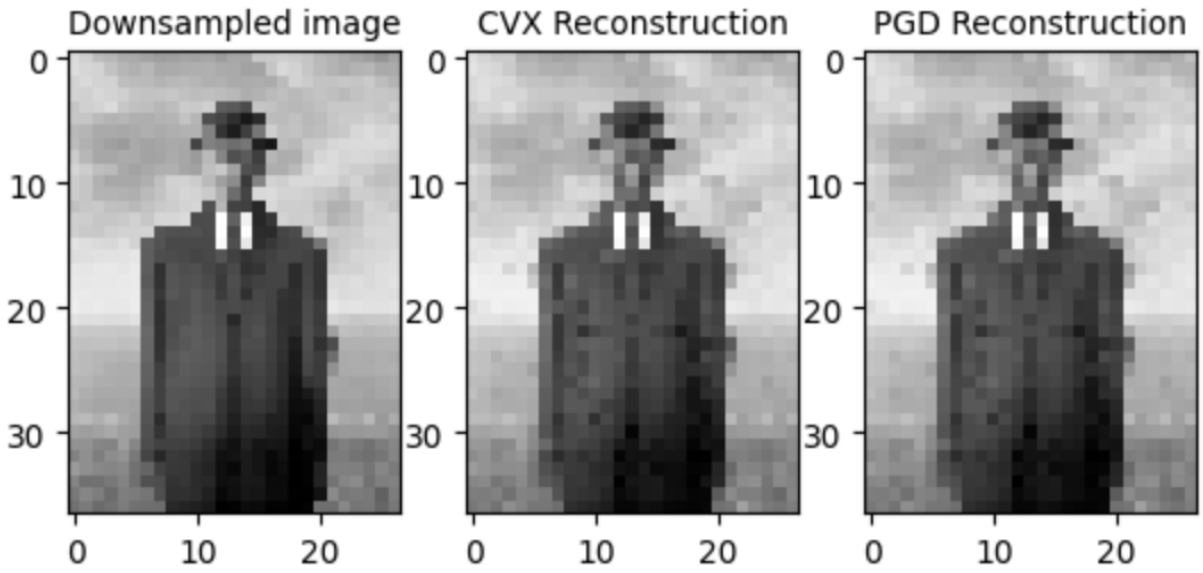


FIGURE 2. Reconstruction of down sampled image using CVX and PGD implementations

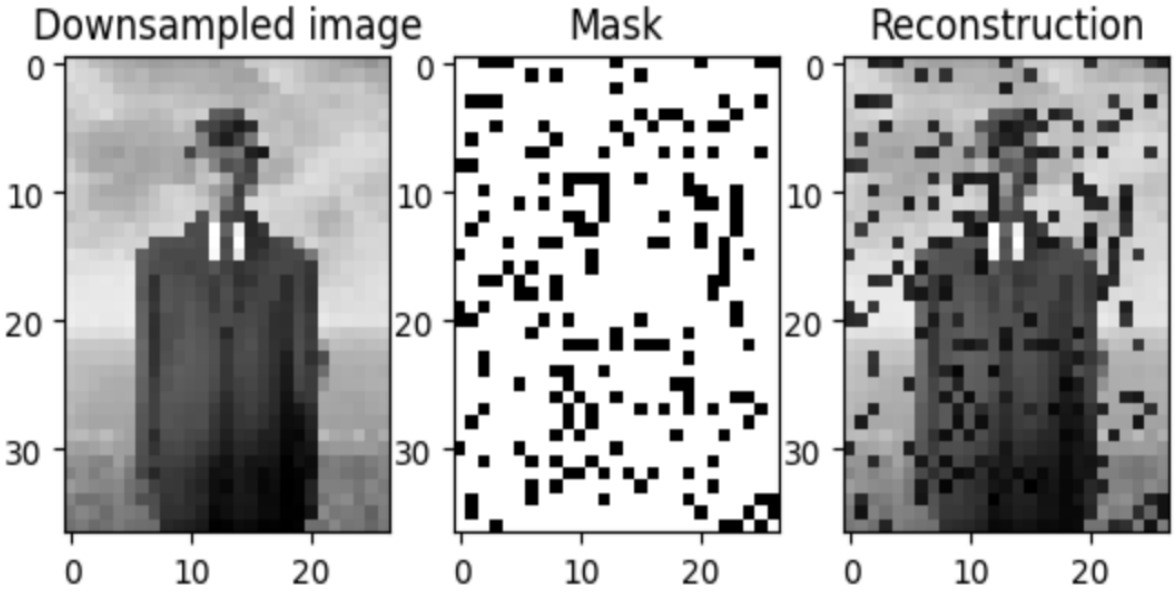


FIGURE 3. Lower regularization parameter causes the mask to appear distinctively on the image.

3.5. **Improvement on PGD.** I also implemented an improved version of PGD - *Fast Iterative Shrinkage-Thresholding algorithm (FISTA)*. This algorithm does not rely only on the last iterate's value, but mixes up with past iterates value to generate a momentum step. This reduces the number of steps required to almost half of what is required for PGD.

The reconstructed original image using PGD and FISTA is displayed 4. The comparison of different algorithms in terms of efficiency and accuracy is tabulated in 1.

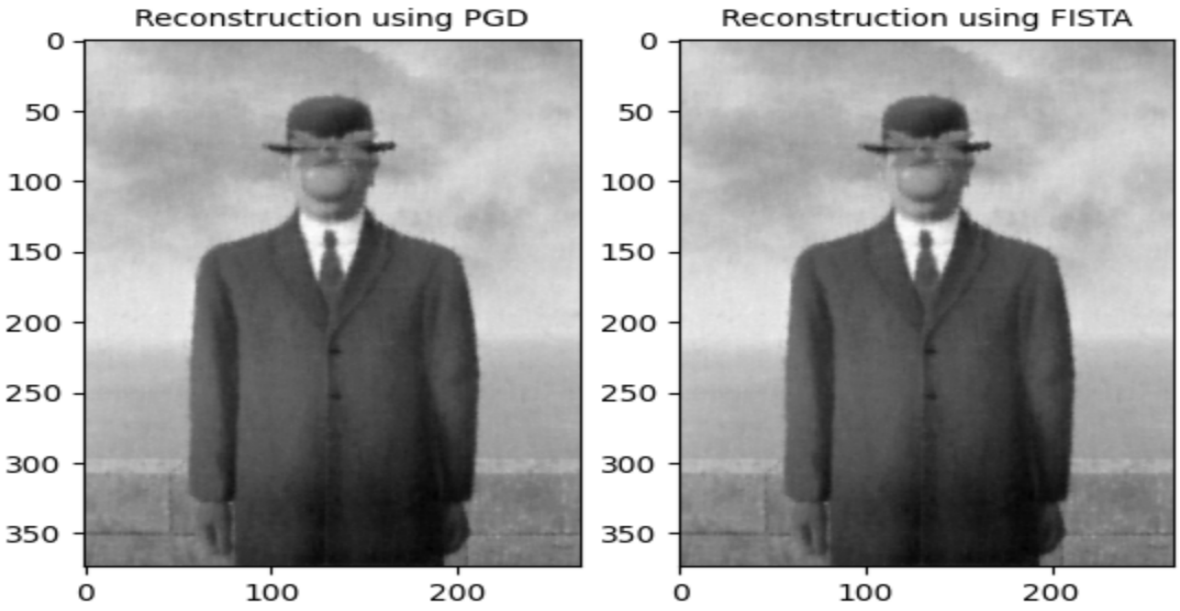


FIGURE 4. Reconstruction of original image using PGD and FISTA implementations

Algorithm	Convergence Steps	Run time	MSE	PSNR
CVXPY	NA	2.75s	0.048	12.81
PGD w/ downscale image	501	0.04s	0.048	12.79
PGD w/ original image	658	1.12s	0.009	20.23
FISTA w/ original image	329	0.59s	0.009	20.23

TABLE 1. Comparison of convergence steps, run times and performance metrics of different algorithms

4. Acknowledgments

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